

Task 2: Normalization

1. Identify normalization issues (1NF, 2NF, 3NF violations):

- 1NF violations:
 - Repeating groups:
 - author_names contain multiple authors violates atomicity.
 - categories contain multiple categories violates atomicity.
 - This table deviates from the requirement that a single cell contains a single value.
 - Fix: Separate rows or related tables should be used to store each author and category.
- 2NF violations:
 - 2NF requires: Every non-key attribute in the 1NF table depends entirely on the complete primary key.
 - If the composite key is (member_id, book_id):
 - member_name depends only on member_id (partial dependency).
 - book_title, author_names, categories, publisher_name, and publisher_country depend only on book_id Partial dependencies exist.
 - Fix: Split into distinct tables for books, borrowing, members, etc.
- 3NF violations:
 - 3NF requires: Table in 2NF and no transitive dependencies.
 - Example: publisher_country depends on publisher_name, not directly on book_id Transitive dependency.
 - Fix: Move publisher info into its own Publisher table.

2. Normalize to 3NF showing each step

- Step 1 – 1NF: Remove repeating groups:

→ Tables after 1NF:

- Member(member_id, member_name)
- Book(book_id, book_title, publisher_name, publisher_country)
- Author(author_id, author_name)
- BookAuthor(book_id, author_id)
- Category(category_id, category_name)
- BookCategory(book_id, category_id)
- Borrow(member_id, book_id, borrow_date, return_date, fine_amount)

- Step 2 – 2NF: Remove partial dependencies

→ Adjustments:

- publisher_name and publisher_country belong to Publisher entity, not Book.
- Book should reference publisher_id, not store publisher details directly.

- Step 3– 3NF: Remove transitive dependencies

→ Create separate Publisher table

- Publisher(publisher_id, publisher_name, publisher_country)

Now Book references Publisher only by publisher_id.

3. Explain the improvements and why each normal form matters

1. 1NF:

- Problem Solved: ensures atomic values and eliminates repetitive groups.
- Why It Matters: Data becomes more consistent and query friendly.

2. 2NF:

- Problem Solved: Eliminates partial dependencies.
- Why It Matters: Each table defines a single entity, preventing redundancy.

3. 3NF:

- Problem Solved: Removes transitive dependencies
- Why It Matters: prevents update abnormalities and enhances data integrity.

4. Show the final normalized tables

1. Member

member_id	member_name
101	John Doe

2. Publisher

publisher_id	publisher_name	publisher_country
1	ABC Press	USA

3. Book

book_id	book_title	publisher_id
B001	The Great Novel	1

4. Author

author_id	author_name
1	Alice Smith
2	Bob Jones

5. BookAuthor

book_id	author_id
B001	1
B001	2

6. Category

category_id	category_name
1	Fiction
2	Drama

7. BookCategory

book_id	category_id
B001	1
B001	2

8. Borrow

borrow_id	member_id	book_id	borrow_date	return_date	fine_amount
1	101	B001	2025-10-01	2025-10-20	5.00

